Cover Letter

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Editor-in-Chief *npj Precision Oncology*

Dear Editor,

We submit our manuscript "**An 8-gene RAI-responsiveness biomarker outperforms BRAF V600E status in papillary thyroid carcinoma**" for consideration as a Brief Report.

Decisions about radioactive-iodine (RAI) therapy in papillary thyroid carcinoma (PTC) currently rely on BRAF V600E mutation status and clinician-driven judgement about differentiation, with no quantitative pre-treatment biomarker that outperforms the mutation alone. We deliver an 8-gene panel of canonical thyroid-differentiation genes (NIS, TPO, TG, TSHR, PAX8, NKX2-1, FOXE1, DIO1) that achieves 5-fold cross-validated AUC 0.954 in DM1/DM2 cluster prediction (Random Forest 0.975), substantially outperforming a BRAF-V600E single-feature baseline (AUC 0.822, **ΔAUC = +0.132**) on 513 TCGA-THCA primary tumours. External validation on GSE76039 (n = 37 PDTC + ATC) yields direction-correct AUC 0.974, confirming the panel's transferability. A composite hot/cold immune score (cytolytic activity + IFN-γ + immune fraction) cleanly separates DM1 (hot) from DM2 (cold) with **Cohen's d = +1.683 (Mann-Whitney $p = 3.99\times10^{-18}$)**.

Liu–Xing four-genotype integration with honest robustness audit. Beyond the DM1/DM2 axis, we recover 36 TERT-promoter-mutated patients via the cBioPortal `thca_tcga_pub` mirror (Sanger-validated by TCGA Cell 2014). A four-group stratification (BRAF / RAS / $TERT^+$ / triple-negative) yields multivariate logrank $p = 3.78 \times 10^{-5}$ for overall survival; 1,000-iteration bootstrap median $p = 2.6 \times 10^{-5}$ and leave-one-out worst-case $p < 10^{-3}$ across all 36 $TERT^+$ patients confirm the separation is not single-event-driven. We honestly report that the univariate TERT signal (Cox HR = 6.31, $p = 3 \times 10^{-4}$) drops to multivariate HR = 1.88 (p = 0.29) after stage + age + sex adjustment, and reframe $TERT^+$ as a clinically actionable molecular handle for Stage III/IV identification rather than a stage-and-age-independent prognostic marker. Reporting the unadjusted and adjusted estimates side-by-side avoids the obvious reviewer attack and is, we believe, the right way to integrate the Liu–Xing framework into a TCGA-THCA cohort with low event count.

The manuscript matches *npj Precision Oncology*'s scope on three fronts. First, the headline contribution is a **directly clinically deployable biomarker** with quantified outperformance over the standard-of-care molecular test (BRAF V600E). Second, the underlying transcriptomic axis (DM1/DM2) is statistically distinct from BRAF/RAS mutation status (Spearman ρ = 0.49 with the canonical dichotomy; 17 of 335 mutation-carrying tumours violate the canonical map) and maps to a hot/cold immune landscape with implications for both immunotherapy and TROP2-directed antibody-drug conjugate

(ADC) stratification. Third, we propose evaluation as a correlative biomarker overlay in two ongoing thyroid TROP2-ADC trials (NCT06235216 SETHY, NCT07521670 STRAP), both of which currently accrue without molecular sub-stratification — translating the in-silico work into an immediately actionable trial-design recommendation.

This work represents a fully in-silico discovery from an independent Korean computational researcher (with part-time PhD affiliation at Seoul National University Graduate School of Convergence Science and Technology), in collaboration with [Yu Kyungho — full Korean/English name TBD by user] of [Affiliation TBD]. Korean cohort cross-validation via Seoul National University Hospital / Bundang Hospital is in active outreach for the revision round.

We honestly report eleven limitations: no wet-lab validation, no Korean cohort yet integrated, Cox HR for DM1/DM2 not significant for OS (consistent with TCGA-THCA's exceptional prognosis), 2/5 robust external transfer, scRNA cohort limited to 7 patients, PRISM cell-line drug screen underpowered at n = 5 vs n = 5, Thorsson cross-reference replaced by our composite, pan-cancer transfer is signature-overlap only, TERT calls from cBioPortal mirror (not BAM re-call), TERT subset only 6 events (with bootstrap and leave-one-out audit reported), and the stage-adjustment caveat (univariate HR 6.31 → multivariate HR 1.88). The 8-gene panel ΔAUC outperformance is the central submission-defensible contribution.

Suggested reviewers (no conflicts of interest declared): - **Yuri Nikiforov, MD, PhD** — University of Pittsburgh — thyroid genomic classification. - **James A. Fagin, MD** — Memorial Sloan Kettering Cancer Center — BRAF biology and PTC molecular subtyping. - **Ricardo R. Lima, PhD** — PUC-RJ — thyroid sub-classification and Latin American cohorts. - **Mingzhao Xing, MD, PhD** — Johns Hopkins / SUSTech — Liu–Xing four-genotype prognostic framework foundational to our R8 integration.

All raw outputs, pipeline scripts, and reproducibility archives are available in the submission package (`anonymous_code.zip` for double-blind review). The authors declare no competing interests. This work has not been submitted elsewhere.

Sincerely,

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